MILITARY SPECIFICATION

PROJECTILE, 120MM, APFSDS, XM829
METAL PARTS ASSEMBLY

This specification is approved for use by the U.S. Army Armament Research and Development Command (ARRADCOM), and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification contains requirements not covered by the drawings and provides quality assurance provisions for the metal parts assembly of one type of armor piercing, fin stabilized, discarding sabot, projectile, designated XM829. This specification is for procurement of hardware for the DTI testing of the XM829.

2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-48078 - Ammunition, Standard Quality Assurance Provisions, General Specification for

STANDARDS

MILITARY

MIL-STD-129 - Marking for Shipment and Storage

MIL-STD-1167 - Ammunition Data Card

MIL-STD-1169 - Packaging, Packing and Marking for Inert Ammunition Components

FSC:1315

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, US Army Armament Research and Development Command, Attn. DRDAR-QA, Dover, New Jersey 07801 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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DRAWINGS

ARRADCOM

12525500 - Cartridge, 120MM, APFSDS-T, XM829
12525501 - Projectile, 120MM, APFSDS, XM829
12525502 - Projectile, Sub-Assembly, XM829
12525503 - Sub-Projectile Assembly
12525504 - Penetrator, Windshield & Tip Assembly
12525505 - Sabot
12525506 - Penetrator, 70 Pressure Flanks
12525507 - Fin
12525508 - Obturator
12525509 - Windshield
12525510 - Tip, Windshield
12525511 - Ring, Tipping
12525512 - Band, Locating, Split
12525513 - Band, Centering
9312761 - Hose Clamp

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

AMERICAN SOCIETY FOR TESTING AND MATERIAL (ASTM)

ASTM D-624	Tear Resistance of Vulcanized Rubber, Method of Test for
ASTM D-785	Rockwell Hardness of Plastics, Test Method for
ASTM D-792	Specific Gravity and Density of Plastics by Displacement, Standard Method of Test for
ASTM D-2240	Indentation Hardness of Rubber and Plastics by Means of Durometer, Method of Test for
ASTM D-2290	Apparent Tensile Strength of Ring or Tubular Plastics by Split Disk Method, Method of Test for
ASTM B-244	Method for Measurement of Thickness of Anodic Coatings on Aluminum with Eddy Current

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103).

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AMERICAL SOCIETY FOR METALS

Metals Handbook, 8th Edition, Volume 7, Atlas of Microstructure of Industrial Alloys.

(Application for copies should be addressed to American Society of Metals, Metals Park, Ohio 44073).

(Technical society and technical association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal Agencies).

3. REQUIREMENTS

3.1 Materials. Materials and parts shall be in accordance with applicable specifications and drawings. Identity of the materials used in manufacturing the projectiles shall be supplied as part of the final inspection data to be submitted to the project officer.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified in Paragraph 2.1 drawings and associated drawings, and with all requirements specified in applicable specifications and standards.

3.3 Protective Coatings.

3.3.1 Hard anodic coating for sabots and fins.

3.3.1.1 Corrosion resistance. The protective finish shall protect the substrate metal so that no evidence of corrosive pitting is found following exposure to salt spray.

3.3.1.2 Abrasion resistance. The hard coating shall have a hard, abrasion resistant finish.

3.3.1.3 Base metal. The base metal shall be free from surface defects caused by machining, cutting, scratching, polishing, buffing, roughing, bending, stretching, deforming, rolling, sandblasting, vaporblasting, etching, inclusions, etc., that will be detrimental to the functional use of the coating. It shall be subjected to such cleaning, etching, anodizing and sealing procedures of the hardcoat process as are necessary to yield coatings that meet all requirements.

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3.3.1.4 Appearance. The anodic coating shall be continuous, smooth, adherent, uniform in appearance and shall be free from powdery areas, loose films, discontinuities such as breaks, scratches, burned areas or other damage. The size and number of contact marks shall be at a minimum consistent with good practice. The location of contact marks shall be in areas of minimum exposure to service environmental conditions. Coated surfaces shall not have a sooty appearance or the presence of a moire pattern.

3.3.1.5 Color. Coatings may vary in color but shall be substantially uniform for parts manufactured from the same alloy and processed to the same nominal coating thickness.

3.4 Proving ground functional tests. The projectiles, when assembled as cartridges and fired from a 120MM, XM256 Gun, shall comply with the following requirements.

3.4.1 Metal parts security. There shall be no evidence during any tests, of metal parts failure in the gun bore or in flight, prior to impact, except as will normally occur with respect to the discarding of the sabot.

3.5 Serial numbers. Each projectile shall be identified by a serial number; the individual projectile numbers shall be consecutively marked. Serial numbers shall be recorded and associated with appropriate test data.

3.6 Data requirements. The contractor shall generate data in accordance with the requirements of the data item descriptions cited in 4.5. The contractor shall maintain inspection records as indicated in 6.3 and submit data as required.

3.7 Workmanship. The requirements for workmanship are as specified on the applicable drawings, referenced specifications and the below listed paragraphs.

3.7.1 Molded components. Molded components shall be free of porosity, laminations and excess flash.

3.7.2 Metal defects. All components shall be free from cracks, seams, splits, cold shuts, inclusions and any other similar defects.

3.7.3 Burrs. No completed component shall have a burr which might interfere with the assembly or function of the item or which might be injurious to personnel handling the item.

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3.7.4 Foreign matter. All parts and assemblies shall be free of dirt, chips, oil, rust, grease, corrosion or other foreign matter. The cleaning method shall not be injurious to any of the parts, nor shall the parts be contaminated by the cleaning agents.

3.7.5 Painting and marking. The painting, protective coating and marking shall be repaired by the assembling contractor on any components when such paintings, protective coatings and markings have been marred or obliterated. (Bare spots in painting and protective coatings, other than marking, are allowable provided they total not more than 0.25 square inch (1.61 square centimeters (cm)) in aggregate, and that the maximum (max.) dimension of any individual bare spot does not exceed 0.18 inch (0.46cm).

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection and standard quality assurance provisions. Unless otherwise specified herein or in the contract, the provisions of MIL-A-48078 shall apply and are hereby made a part of this detail specification.

4.2 Classification of inspection. Quality conformance inspections shall be conducted on this item.

4.3 Inspection conditions. Unless otherwise specified in the classification of defects and tests tables, sampling plans and procedures for major and minor defects shall be in accordance with MIL-STD-105, Inspection Level II.

4.4 Quality conformance inspection.

4.4.1 Lot formation. Lots shall comply with the lot formation provisions of MIL-A-48078.

4.4.2 Examination and testing. See MIL-A-48078. Variables data shall be recorded, on a sampled basis, for every dimension indicated in this section under the column titled "Paragraph reference/inspection method". Gage, visual or special inspections shall be interpreted for use in only this contract, as that inspection technique, approved by the Government, that will yield accurate and precise information for the minimum cost to the Government. Specifically, the word "gage" is not meant to imply the mandatory use of special purpose, dedicated gages during subject contract, but is meant to allow any government approved technique which insures compliance with the drawings.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET	OF	DRAWING NUMBER
4.4.2.1	Centering Band	1	1	12525513

NEXT HIGHER ASSEMBLY: 12525502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Inside diameter (Before Splitting)		0.40%	3.2	Gage
Minor					
201	Wall variation		0.65%	3.2	Gage
202	Cut location, 4 places		0.65%	3.2	Gage
203	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE			SHEET 1 OF 1	DRAWING NUMBER 12525512
4.4.2.2	Band, Locating, Split				NEXT HIGHER ASSEMBLY 12525502
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length (before cutting)		0.40%	3.2	Gage
102	Absence of inside edge angle (before cutting)		0.40%	3.2	Visual
103	Water absorption & content	a.		3.2	4.5.1
104	Material hardness	a.		3.2	4.5.2
Minor					
201	Material specific gravity	a.		3.2	4.5.3
202	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

a. The sample size for each lot of material shall be 5 each.

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CLASSIFICATION OF DEFECTS & TESTS

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.3	Obturator		12525508
			NEXT HIGHER ASSEMBLY 12525502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length		0.40%	3.2	Gage
102	Forward obturator diameter		0.40%	3 2	Gage
103	Outside taper angle		0.40%	3.2	Gage
104	Outside rear diameter		0.40%	3.2	Gage
105	Diameter at bottom of outside notch		0.40%	3.2	Gage
106	Length of outside notch		0.40%	3.2	Gage
107	Inside diameter at rear of obturator		0.40%	3.2	Gage
108	Water absorption & moisture content	a.		3.2	4.5.4
109	Hardness	a.		3.2	4.5.5
110	Specific gravity	a.		3.2	4.5.6
111	Angle of both saw teeth		0.40%	3.2	Gage
112	Length to first tooth basic diameter		0.40%	3.2	Gage
113	Length to end of first tooth		0.40%	3.2	Gage
Minor					
201	Length to second tooth basic diameter		0.65%	3.2	Gage
202	Absence of three O.D. holes		0.65%	3.2	Visual
203	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES: a. The sample size for each lot of material shall be 5 each.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.4	Windshield Tip		12525510
			NEXT HIGHER ASSEMBLY
			12525504

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Location of aft basic diameter		0.40%	3.2	Gage
103	Location of forward basic diameter		0.40%	3.2	Gage
104	Thread pitch diameter, maximum		0.40%	3.2	Gage
105	Thread minor diameter, maximum		0.40%	3.2	Gage
106	Aft end diameter		0.40%	3.2	Gage
107	Aft face perpendicularity		0.40%	3.2	Gage
Minor					
201	Protective finish		0.65%	3.2	Visual
202	Evidence of poor workmanship		0.65%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.5	Windshield		12525509
			NEXT HIGHER ASSEMBLY
			12525504

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length from aft end to shoulder		0.40%	3.2	Gage
102	Diameter to shoulder perpendicularity		0.40%	3.2	Gage, Variables Data
103	Cavity depth		0.40%	3.2	Gage
104	(Reserved)				
105	Smallest cavity diameter		0.40%	3.2	Gage
106	Diameter at inside rear locating point		0.40%	3.2	Gage
107	True position of inside rear diameter		0.40%	3.2	Gage
108	O.D. at aft outside rear locating point		0.40%	3.2	Gage
109	Runout of O.D.		0.40%	3.2	Gage
110	O.D. at 3rd forward outside locating point		0.40%	3.2	Gage
111	Runout of 3rd forward outside O.D.		0.40%	3.2	Gage
112	Thread pitch diameter, minimum		0.40%	3.2	Gage
113	Thread major diameter, minimum		0.40%	3.2	Gage
114	True position thread pitch diameter		0.40%	3.2	Gage
115	Absence of thru hole		0.40%	3.2	Gage
116	Diameter at front end		0.40%	3.2	Gage
117	O.D. at 1st forward locating point		0.40%	3.2	Gage
118	O.D. at 2nd forward locating point		0.40%	3.2	Gage
119	Runout at 2nd forward O.D.		0.40%	3.2	Gage
120	Absence of rear counterbore angle		0.40%	3.2	Gage
121	Rearmost I.D.		0.40%	3.2	Gage
Minor					
201	Protective finish		100%	3.2	Visual
202	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.6	Split Band, Rear				12525512
					NEXT HIGHER ASSEMBLY
					12525502
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Length (before cutting)		0.40%	3.2	Gage
102	Absence of inside edge angle (before cutting)		0.40%	3.2	Visual
103	Water absorption & content	a.		3.2	4.5.1
104	Material hardness	a.		3.2	4.5.2
Minor					
201	Material specific gravity	a.		3.2	4.5.3
202	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

a. The sample size for each lot of material shall be 5 each.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE		SHEET 1 OF 2		DRAWING NUMBER
4.4.2.7	Fin				12525507
					NEXT HIGHER ASSEMBLY
					12525503
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Hub length		0.40%	3.2	Gage
103	Fin edge diameter location		0.40%	3.2	Gage
104	True position fin leading edge diameter		0.40%	3.2	Gage
105	Forward cavity I.D.		0.40%	3.2	Gage
106	True position of forward I.D.		0.40%	3.2	Gage
107	Forward unthreaded hub depth		0.40%	3.2	Gage
108	(Reserved)				
109	Forward cavity depth		0.40%	3.2	Gage, Variables Data
110	Thread pitch diameter, maximum		0.40%	3.2	Gage
111	Thread minor diameter, maximum		0.40%	3.2	Gage
112	Leading edge angle		0.40%	3.2	Gage
113	Tracer cavity depth		0.40%	3.2	Gage
114	Full thread, minimum		0.40%	3.2	Gage
115	Tracer cavity thread pitch diameter, maximum		0.40%	3.2	Gage
116	Tracer cavity thread minor diameter, maximum		0.40%	3.2	Gage
117	Tracer cavity diameter		0.40%	3.2	Gage
118	True position of tracer cavity		0.40%	3.2	Gage
119	Aft fin angle		0.40%	3.2	Gage
120	Outer fin edge diameter		0.40%	3.2	Gage, Variables Data
121	True position of fin edge diameter		0.40%	3.2	Gage
122	Inside bottom edge diameter		0.40%	3.2	Gage
123	Hub diameter		0.40%	3.2	Gage
124	True position of fin blades		0.40%	3.2	Gage
125	Blade trailing edge angle		0.40%	3.2	Gage

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 2 OF 2	DRAWING NUMBER
4.4.2.7	Fin, (continuation sheet)		12525507
			NEXT HIGHER ASSEMBLY
			12525503

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
126	Blade leading edge angle		0.40%	3.2	Gage
127	Thickness of leading edge		0.40%	3.2	Gage
128	Outer blade location thickness		0.40%	3.2	Gage
129	Second outer blade location thickness		0.40%	3.2	Gage
130	Second inner blade location thickness		0.40%	3.2	Gage
131	Inner blade location thickness		0.40%	3.2	Gage
Minor					
201	Absence of edge blend radii		0.65%	3.2	Visual
202	Evidence of poor workmanship		100%	3.7	Visual

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CLASSIFICATION OF DEFECTS & TESTS

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.8	Penetrator, Windshield & Tip Assembly		12525504
			NEXT HIGHER ASSEMBLY
			12525503

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum length		0.40%	3.2	Gage
102	Runout of windshield tip		0.40%	3.2	Gage, Variables Data
103	Presence of windshield sealant		0.40%	3.2	Gage
Minor					
201	Evidence of poor workmanship		100%	3.7	Visual

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PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.9	Ring, Tipping		12525511
			NEXT HIGHER ASSEMBLY
			12525501

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Outside diameter		0.40%	3.2	Gage
103	Aft inside diameter		0.40%	3.2	Gage
104	Forward inside diameter		0.40%	3.2	Gage
105	Depth of forward inside diameter		0.40%	3.2	Gage
106	Runout of forward inside diameter		0.40%	3.2	Gage
107	Depth of cuts at aft face (4 each)		0.40%	3.2	Gage
108	Length from bottom of forward cut to aft face (4 each)		0.40%	3.2	Gage
109	True position of 4 cuts		0.40%	3.2	Gage
Minor					
201	Chamfer on aft inside diameter		0.65%	3.2	Gage
202	Surface Finish		0.65%	3.2	Gage
203	Evidence of poor workmanship		100%	3.7	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 1 OF 3	DRAWING NUMBER
4.4.2.10	Sabot		12525505
			NEXT HIGHER ASSEMBLY
			12525502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Overall length		0.40%	3.2	Gage
102	Diameter at forward scoop datum length		0.40%	3.2	Gage
103	Forward ramp front basic location		0.40%	3.2	Gage
104	Forward ramp aft basic location		0.40%	3.2	Gage
105	First saddle diameter at basic length		0.40%	3.2	Gage
106	Forward scoop, outside diameter		0.40%	3.2	Gage
107	(Reserved)				
108	Length from aft end to obturator forward edge		0.40%	3.2	Gage
109	Forward scoop to front end distance		0.40%	3.2	Gage
110	Locating band slot width		0.40%	3.2	Gage
111	Aft end to locating band slot length		0.40%	3.2	Gage, Variables Data
112	Length, aft end to obturator aft end		0.40%	3.2	Gage, Variables Data
113	Aft end diameter		0.40%	3.2	Gage
114	Rear bore-rider diameter		0.40%	3.2	Gage
115	Runout of bore-rider O.D.		0.40	3.2	Gage, Variables Data
116	Diameter of aft ramp at rearmost basic		0.40%	3.2	Gage
117	Runout of rearmost diameter		0.40%	3.2	Gage
118	Diameter of aft ramp at 2nd rearmost basic		0.40%	3.2	Gage
119	Runout of second rearmost diameter		0.40%	3.2	Gage
120	Diameter of aft ramp at 3rd rearmost basic		0.40%	3.2	Gage
121	Runout of 3rd rearmost diameter		0.40%	3.2	Gage
122	Diameter of aft ramp at 4th rearmost basic		0.40%	3.2	Gage

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 2 OF 3	DRAWING NUMBER
4.4.2.10	Sabot (continuation sheet)		12525505
			NEXT HIGHER ASSEMBLY 12525502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
123	Runout of 4th rearmost diameter		0.40%	3.2	Gage
124	Diameter in locating band slot		0.40%	3.2	Gage
125	Runout of slot diameter		0.40%	3.2	Gage
126	Angle on first obturator tooth		0.40%	3.2	Gage
127	Angle on second obturator tooth		0.40%	3.2	Gage
128	(Reserved)				
129	Location of second tooth basic diameter		0.40%	3.2	Gage
130	Location of first tooth basic diameter		0.40%	3.2	Gage
131	Width of second obturator tooth		0.40%	3.2	Gage
132	Sabot segment angle, 4 each		0.40%	3.2	Gage
133	Pitch diameter (note b)		0.40%	3.2	Gage/Special
134	Profile of buttress grooves (see notes a & b)		0.40%	3.2	Gage/Special
135	Friction foot diameter		0.40%	3.2	Gage
136	True position of friction foot		0.40%	3.2	Gage
137	Diameter in centering band slot		0.40%	3.2	Gage
138	Length forward edge to dovetail bottom		0.40%	3.2	Gage
139	Absence of forward angle		0.40%	3.2	Visual
140	Length forward edge to rear dovetail bottom		0.40%	3.2	Gage
141	Length forward edge to aft edge		0.40%	3.2	Gage
142	Microstructure (note c)			3.2/3.7	4.5.7
143	Protective finish		100%	3.2	Visual
144	Presence of protective finish in friction foot		100%	3.2	Visual
Minor					
201	Obturator aft end radius		0.65%	3.2	Gage

NOTES: a. Profile includes inspection of pressure flank angle, clearance flank angle, groove minor diameter radius and radius of front & rear surface of groove. b. Check at first, last & every 10th groove in between. c. One bar from ea. solution heat treatment batch.

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 3 OF 3	DRAWING NUMBER
4.4.2.10	Sabot (continuation sheet)		12525505
			NEXT HIGHER ASSEMBLY
			12525502

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
202	Absence of locating band & obturator bottom radius		0.65%	3.2	Visual
203	Absence of dovetail bottom radius		0.65%	3.2	Visual
204	Marking missing or incomplete		100%	3.2	Visual
205	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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QUALITY CONFORMANCE INSPECTION

CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.11	Sub-projectile Assembly		12525503
			NEXT HIGHER ASSEMBLY: 12525501

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum length		0.40%	3.2	Gage
102	Fin torque		0.40%	3.2	Gage
103	Fin runout		0.40%	3.2	Gage, Variables Date
Minor					
201	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE	SHEET 1 OF 1	DRAWING NUMBER
4.4.2.12	Projectile, sub-assembly, XM829		12525502
			NEXT HIGHER ASSEMBLY 12525501

CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	Maximum diameter at centering band		100%	3.2	Gage
102	Runout of centering band diameter		0.40%	3.2	Gage
103	Maximum diameter at boreriding surface		100%	3.2	Gage
104	Runout of boreriding surface		0.40%	3.2	Gage
105	Diameter at rear ramp point		0.40%	3.2	Gage
106	Diameter at aft seal		0.40%	3.2	Gage
107	Absence of aft seal end chamfer		0.40%	3.2	Visual
108	Segment separation distance, 4 each		0.40%	3.2	Visual
109	Diameter at obturator basic length		0.40%	3.2	Gage, Variables Data
110	Maximum gap between sabot segments		100%	3.2	Gage
Minor					
201	Sabot segments assembled incorrectly		100%	3.2	Visual
202	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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CLASSIFICATION OF DEFECTS & TESTS

MIL-P-63510(AR)

PARAGRAPH	TITLE		SHEET 1 OF 1		DRAWING NUMBER
4.4.2.13	Projectile, 120MM, APFSDS, XM829				12525501
					NEXT HIGHER ASSEMBLY
					12525500
CATEGORY	EXAMINATION OR TEST	NO. OF SAMPLE UNITS	AQL OR 100%	REQUIREMENT PARAGRAPH	PARAGRAPH REFERENCE /INSPECTION METHOD
Critical	None defined				
Major					
101	(Reserved)				
102	Bourrelet diameter at centering band		0.40%	3.2	Gage
103	Boreriding diameter at locating band		0.40%	3.2	Gage
Minor					
201	Evidence of excess fin sealant		100%	3.2	Visual
202	Unit weight (information)		100%	3.2	Gage, Variables Data
203	Marking missing or incomplete		100%	3.4	Visual
204	Any fin blade bent		100%	3.2	Gage
205	Evidence of poor workmanship		100%	3.7	Visual

NOTES:

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4.4.3 Testing. Testing is called out in the Quality Conformance Inspection Tables.

4.4.4 Inspection equipment. The inspection equipment required to perform the examinations and tests prescribed herein are described in the Paragraph Reference/Inspection Method column in the tables starting with paragraph 4.4.2.1. The contractor shall submit for approval, inspection equipment designs in accordance with the terms of the contract. See Section 6 of MIL-A-48078 and 6.2 herein. Submission shall be in accordance with DI-R-1714.

4.5 Test methods and procedures.

4.5.1 Split band moisture content and water absorption. All specimens shall be wiped clean of any loose material with a lint free cloth or tissue, and weighed on an analytical balance. This is the initial weight. The specimens are then placed in an oven maintained at 160°F ± 10°F for approximately 24 hours. Cool in a desiccator and reweigh. After initial conditioning, condition for approximately 8 hours, cool in a desiccator and reweigh. Repeat oven conditioning and cooling until constant weight is achieved. Constant weight is achieved when the difference between two consecutive weighings is no more than 5 milligrams. This is the conditioned weight. Completely immerse the dried specimens in a container of distilled water maintained at 73°F ± 8°F for a period of 24 hours, minus 0 plus 1/2 hour. Specimens shall be individually removed from the water, all surface water wiped off with a dry, lint free cloth, and weighed immediately on an analytical balance. This gives the wet weight. Percent moisture content shall be calculated as follows:

$$\frac{\text{Initial weight - conditioned weight}}{\text{Initial weight}} \times 100 = \%\ \text{Moisture Content}$$

Percent water absorption shall be calculated as follows:

$$\frac{\text{Wet weight - conditioned weight}}{\text{Conditioned weight}} \times 100 = \%\ \text{Water Absorption}$$

Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

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4.5.2 Split band hardness. Rockwell hardness determination shall be made in accordance with ASTM D-785 using the R scale. Four readings 90° apart shall be taken on the outside of the obturator. Failure of any reading to meet the applicable requirement shall be cause for rejection of the lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

4.5.3 Split band specific gravity. Two specimens shall be selected from each band for use in determining the specific gravity. The test shall be performed and the specimen size shall be selected in accordance with ASTM Method D-792, Method A-1. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

4.5.4 Obturator moisture content and water absorption. All specimens shall be wiped clean of any loose material with a lint free cloth or tissue, and weighed on an analytical balance. This is the initial weight. The specimens are then placed in an oven maintained at 160°F ± 10°F for approximately 24 hours. Cool in a desiccator and reweigh. After initial conditioning, condition for approximately 8 yours, cool in a desiccator and reweigh. Repeat oven conditioning and cooling until constant weight is achieved. Constant weight is achieved when the difference between two consecutive weighings is no more than 5 miligrams. This is the conditioned weight. Completely immerse the dried specimens in a container of distilled water maintained at 73°F ± 8°F for a period of 24 hours, minus 0 plus 1/2 hour. Specimens shall be individually removed from the water, all surface water wiped off with a dry, lint free cloth, and weighed immediately on an analytical balance. This gives the wet weight. Percent moisture content shall be calculated as follows:

$$\frac{\text{Initial weight - conditioned weight}}{\text{Initial weight}} \times 100 = \%\ \text{Moisture Content}$$

Percent water absorption shall be calculated as follows:

$$\frac{\text{Wet weight - conditioned weight}}{\text{Conditioned weight}} \times 100 = \%\ \text{Water Absorption}$$

Failure of any specimen to meet the applicable requirement shall be cause for rejection of the Lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

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4.5.5 Obturator hardness. Rockwell hardness determination shall be made in accordance with ASTM D-785 using the R scale. Four readings 90° apart shall be taken on the outside of the obturator. Failure of any reading to meet the applicable requirement shall be cause for rejection of the lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

4.5.6 Obturator specific gravity. Two specimens shall be selected from each band for use in determining the specific gravity. The test shall be performed and the specimen size shall be selected in accordance with ASTM Method D-792, Method A-1. Failure of any specimen to meet the applicable requirement shall be cause for rejection of the lot. Results of this inspection shall be submitted in accordance with DI-T-1906.

4.5.7 Examination of sabot microstructure. Prior to machining, one extruded bar from each solution heat treatment batch shall be selected for this test. A wafer approximately 0.50 inches thick, shall be removed from each end of the bar stock and a longitudinal section, a minimum of 0.25 inches in length, shall be cut from the wafer to be prepared for metallographic examination. When suitable prepared, the specimen shall be etched with Keller's reagent and examined by scanning under a microscope at a minimum magnification of 500 diameters. Overheating shall be evidenced by the presence of rosettes in the microstructure or incipient melting at the grain boundries. Microstructure examination of specimens shall be performed by a skilled metallographer familiar with aluminum alloys. Rosettes are defined by figure 2018, page 245 and incipient melting of the grain boundries by figure 2073, page 251 of the American Society for Metals, Metal Handbook, 8th Edition. Results of this test shall be submitted in accordance with DI-T-1906.

5. PREPARATION FOR DELIVERY

5.1 Packing and packaging. Preservation, packaging, and packing of the projectile are to be conducted and performed per applicable Nuclear Regulatory Commission (NRC), Department of Energy (DOE), and Department of Transportation (DOT) regulations as described in the contractor's staballoy NRC handling license.

5.2 Marking. Marking of the projectile shall be in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. The projectiles covered by the Quality Evaluation Plan are intended for use in the cartridge, 120MM, APFSDS-T, XM829.

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6.2 Submission of designs for approval. See MIL-A-48078. Submit equipment designs as required to:

Commander
US Army Armament Research & Development Command
ATTN: DRDAR-QAT-I
Dover, New Jersey 07801

6.3 Data requirements. When this specification is used in a procurement which incorporates DD Form 1423 and invokes the provision of 7-104.9(n) of the Armed Services Procurement Regulations, the data requirements identified below will be developed as specified by an approved Data Item Description (DD Form 1664) and delivered in accordance with the approved Contract Data Requirements List (DD Form 1423) incorporated into the contract. When the provisions of ASPR-7-104.9(n) are not invoked, the data specified below will be delivered by the contractor in accordance with the contract requirements. Deliverable data required by this specification is cited in the following paragraphs:

PARAGRAPH	DATA REQUIREMENT	APPLICABLE DID
4.4.4	Quality Engineering Acceptance Inspection Equipment Description Documentation	DI-R-1714
4.5.1	Test & Demo. Reports	DI-T-1906
4.5.2	Test & Demo. Reports	DI-T-1906
4.5.3	Test & Demo. Reports	DI-T-1906
4.5.4	Test & Demo. Reports	DI-T-1906
4.5.5	Test & Demo. Reports	DI-T-1906
4.5.6	Test & Demo. Reports	DI-T-1906
4.5.7	Test & Demo. Reports	DI-T-1906

(Copies of data item descriptions required by the contractor in connection with specific functions should be obtained from the procuring activity or as directed by the contracting officer.)

6.4 Submission of ammunition data cards. In addition to the normal distribution of records, one copy of all ammunition data cards shall be forwarded to each of the following:

Commander
US Army Armament Research & Development Command
ATTN: DRDAR-QAT-A and DRDAR-LCU-CT
Dover, New Jersey 07801

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6.5 Handling of staballoy. Care shall be taken to ensure that all procedures for handling staballoy as outlined by the NRC, Dept. of Energy (DOE) and the individual contractor's regulations and/or licenses are followed.

CUSTODIAN:
Army - AR

PREPARING ACTIVITY:
Army - AR

Project No. 1315-A421

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APPENDIX A

STANDARD OPERATING PROCEDURE

for

CARTRIDGE, XM829, 120MM, APFSDS-T

LOAD, ASSEMBLE AND PACK (LAP)

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APPENDIX A

TO BE DETERMINED

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APPENDIX B

INSPECTION CRITERIA

for

PENETRATOR, 70 PRESSURE FLANKS

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APPENDIX B

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